

14 2026-02-23 | Week 07 | Lecture 14

14.1 Expectation

Recall that the *expected value* of a discrete random variable is

$$\mu = \mathbb{E}[X] = \sum_x x \cdot p(x)$$

where the x in the summation runs over all possible values of X .

In words, the expected value is the *long-run average*, meaning that if you repeated an experiment many times (independently), the long-run average would converge to $\mathbb{E}[X]$.

Remark 58 (Notation). We sometimes use the notation μ or μ_X for $\mathbb{E}[X]$.

The following is a simple but important fact:

Proposition 59 (Linearity of Expectation). *Let a, b be numbers and X and Y be a random variables. Then*

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

and

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

We saw this earlier, when we looked at the expected value of playing the lottery once ($-\$1$) and of playing 10 times ($-\$10$), etc.

Theorem 60 (Expectation of a function of X). *Let X be a discrete random variable with pmf $p(x)$. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be any function. Then*

$$\mathbb{E}[h(X)] = \sum_x h(x)p(x)$$

provided that the sum on the right hand side is absolutely convergent. As usual, the summation runs over all possible values of X .

Example 61. Suppose Y is a discrete random variable with pmf

$$p(-2) = \mathbb{P}[Y = -2] = 0.1$$

$$p(-1) = \mathbb{P}[Y = -1] = 0.3$$

$$p(1) = \mathbb{P}[Y = 1] = 0.4$$

$$p(2) = \mathbb{P}[Y = 2] = 0.2$$

and such that $\mathbb{P}[Y = y] = 0$ if $y \notin \{-2, -1, 1, 2\}$ Find the following quantities:

(a) $\mathbb{E}[Y]$

(b) $\mathbb{E}[3Y + 7]$

(c) $\mathbb{E}[Y^3 + 2Y]$

(d) $\mathbb{E}[e^X]$

Solution to (a):

$$\begin{aligned}\mathbb{E}[Y] &= -2(0.1) + (-1)(0.3) + (1)(0.4) + (2)(0.2) \\ &= -.2 - .3 + .4 + .4 \\ &= .3\end{aligned}$$

Solution to (b):

$$\begin{aligned}\mathbb{E}[3Y + 7] &= 3\mathbb{E}[Y] + 7 && \text{by Proposition 59} \\ &= 3(0.3) + 7 && \text{by our answer to part (a).} \\ &= 7.9.\end{aligned}$$

Solution to (c): Here we will apply Theorem 60 with $h(x) = x^3 + 2x$:

$$\begin{aligned}\mathbb{E}[Y^3 + 2Y] &= \sum_{y \in \{-2, -1, 1, 2\}} h(y)p(y) \\ &= h(-2)p(-2) + h(-1)p(-1) + h(1)p(1) + h(2)p(2) \\ &= (-12)(0.1) + (-3)(0.3) + (3)(0.4) + (12)(0.2) \\ &= 1.5.\end{aligned}$$

Solution to (d): Here we will apply Theorem 60 with $h(x) = e^x$:

$$\begin{aligned}\mathbb{E}[e^Y] &= \sum_{y \in \{-2, -1, 1, 2\}} h(y)p(y) \\ &= \sum_{y \in \{-2, -1, 1, 2\}} e^y p(y) \\ &= e^{-2}p(-2) + e^{-1}p(-1) + e^1p(1) + e^2p(2) \\ &= e^{-2} \cdot 0.1 + e^{-1} \cdot 0.3 + e^1 \cdot 0.4 + e^2 \cdot 0.2 \\ &\approx 2.689.\end{aligned}$$

End of Example 61. $\square$

14.2 Variance

Definition 62 (Variance). The **variance** of a random variable X is

$$\text{Var}(X) := \mathbb{E}[(X - \mu)^2]$$

where $\mu = \mathbb{E}[X]$. (Note: variance is often denoted by σ^2)

Variance is a measure of how likely the value of a random variable is to be far from its expected value. In an ideal world, we would measure this by

$$\mathbb{E}[|X - \mu|] = (\text{expected distance of } X \text{ from its mean})$$

But unfortunately, the absolute value function is mathematically difficult to work with, so instead we use

$$\mathbb{E}[(X - \mu)^2] = \text{Var}(X)$$

which is mathematically easier to work with because it doesn't have an absolute value.

Example 63. Let X and Y be random variables defined as follows:

$$X = \begin{cases} +1 & : \text{ with probability } \frac{1}{2} \\ -1 & : \text{ with probability } \frac{1}{2} \end{cases} \quad \text{and} \quad Y = \begin{cases} +1000 & : \text{ with probability } \frac{1}{2} \\ -1000 & : \text{ with probability } \frac{1}{2} \end{cases}$$

In both cases, $\mathbb{E}[X] = 0$ and $\mathbb{E}[Y] = 0$. But despite having the same expected value, these are very different random variables: the value of Y deviates from its mean much more than X does. Indeed, direct computation (shown below) shows the variance of Y to be much larger than the variance of X :

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2] && \text{since } \mu = \mathbb{E}[X] = 0 \\ &= (1)^2(.5) + (-1)^2(.5) && \text{by Theorem 60} \\ &= 1.\end{aligned}$$

On the other hand,

$$\begin{aligned}\text{Var}(Y) &= \mathbb{E}[(Y - \mu)^2] \\ &= \mathbb{E}[Y^2] \\ &= (1000)^2(.5) + (-1000)^2(.5) \\ &= 1000000.\end{aligned}$$

since $\mu = \mathbb{E}[Y] = 0$

by Theorem 60

End of Example 63. $\square$